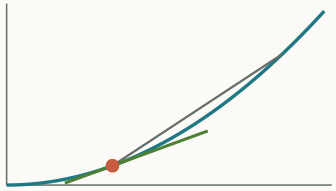
Differentiation CHEAT SHEET

RATE OF CHANGE AND THE DERIVATIVE



chord: average

tangent: instant

the chord's slope → the tangent's

average rate = $\frac{f(b)-f(a)}{b-a}$ (a chord's slope)

$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

notation: $f'(x)$, $\frac{dy}{dx}$, $\frac{d}{dx}f(x)$, y'

TWO SPECIAL LIMITS

$\lim_{h\to 0} \frac{\sin h}{h} = 1$ $\lim_{h\to 0} \frac{\cos h - 1}{h} = 0$

$\lim_{h\to 0} \frac{e^h - 1}{h} = 1$ (h in radians)

sin h < h < tan h squeezes sin h / h to 1

CONTINUITY AND DIFFERENTIABILITY

continuous at a: $\lim_{x\to a} f(x) = f(a)$

differentiable at a: left and right derivatives agree

differentiable ⇒ continuous (not the reverse: |x| at 0)

THE RULES

constant: $\frac{d}{dx}(c) = 0$

power: $\frac{d}{dx}(x^n) = n\,x^{n-1}$

multiple: $(c\,f)' = c\,f'$

sum: $(u \pm v)' = u' \pm v'$

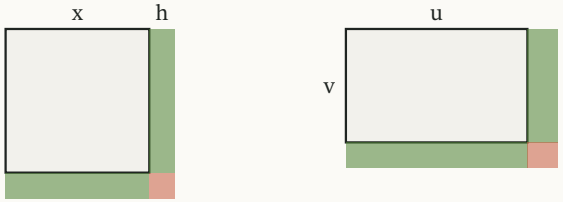
product: $(uv)' = u\,v' + v\,u'$

quotient: $(\frac{u}{v})' = \frac{v\,u' - u\,v'}{v^2}$

chain: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

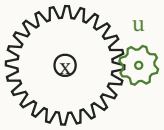
quotient: the bottom goes first · chain: work from the outside in

WHY THE RULES WORK



$(x+h)^2 - x^2 = 2xh + h^2$ $d(uv) = u\,dv + v\,du$

the corner (coral) shrinks far faster than the strips



x turns once, u turns 3 times: du/dx = 3

chain: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

the gears' rates multiply down the chain

FROM FIRST PRINCIPLES

$x^2 \to 2x$ $x^3 \to 3x^2$

$1/x \to -1/x^2$ $\sqrt{x} \to \frac{1}{2\sqrt{x}}$

$ax + b \to a$ $c \to 0$

STANDARD DERIVATIVES

f(x)	f′(x)
x^n	$n\,x^{n-1}$
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$\frac{1}{x}$
$\log_a x$	$\frac{1}{x \ln a}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sin^{-1} x$	$\frac{1}{\sqrt{(1-x^2)}}$
$\cos^{-1} x$	$\frac{-1}{\sqrt{(1-x^2)}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\cot^{-1} x$	$\frac{-1}{1+x^2}$
$\sec^{-1} x$	$\frac{1}{\lvert x\rvert\sqrt{(x^2-1)}}$
$\operatorname{cosec}^{-1} x$	$\frac{-1}{\lvert x\rvert\sqrt{(x^2-1)}}$

every co- function brings a minus (coral) · angles in radians

Differentiation CHEAT SHEET

TECHNIQUES

implicit: differentiate both sides; $\frac{d}{dx}(y^2) = 2y\,\frac{dy}{dx}$

$x^2 + y^2 = 169$ at $(5, 12)$: $\frac{dy}{dx} = -\frac{x}{y} = -\frac{5}{12}$

log diff.: $y = x^x \Rightarrow \ln y = x \ln x \Rightarrow y' = x^x(\ln x + 1)$

parametric: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \div \frac{dx}{dt}$$

higher: $f''(x) = \frac{d^2y}{dx^2}$ (the slope of the slope)

$\sin \to \cos \to -\sin \to -\cos \to \sin$ (repeats every 4)

TANGENTS AND NORMALS

tangent: $y - f(a) = f'(a)(x - a)$

normal: $y - f(a) = -\frac{1}{f'(a)}(x - a)$

$y = x^3 - 2x$ at $x = 2$: tangent $y = 10x - 16$

normal there: $x + 10y = 42$

tangent and normal slopes multiply to -1

LINEAR APPROXIMATION

$f(a + h) \approx f(a) + h\,f'(a)$

$dy = f'(x)\,dx$

$\sqrt{26} \approx 5 + \frac{1}{10} = 5.1$ (true 5.0990…)

only near a: far away the curve bends off its tangent

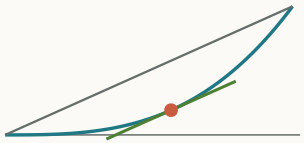
RELATED RATES: FOUR STEPS

1. draw it; name what changes
2. write an equation linking them
3. differentiate with respect to t
4. then put in the moment's numbers

ROLLE AND THE MEAN VALUE THEOREM

Rolle: $f(a) = f(b) \Rightarrow f'(c) = 0$ for some $a < c < b$

mean value: $f'(c) = \frac{f(b) - f(a)}{b - a}$ for some $a < c < b$



tangent parallel to the chord

x^3 on $[0, 2]$: $c = 2/\sqrt{3}$

THE SHAPE OF A GRAPH

$f' > 0$: increasing $f' < 0$: decreasing

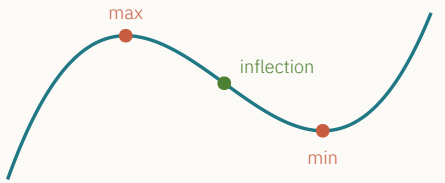
critical points: $f'(c) = 0$ or $f'(c)$ undefined

first test: f' + → − max; − → + min

second test: $f'(c) = 0$, $f''(c) < 0$ max; $f''(c) > 0$ min

$f'' > 0$: concave up (cup) $f'' < 0$: concave down (cap)

inflection: f'' changes sign (x^4 at 0 does not)



$x^3 - 6x^2 + 9x + 1$

OPTIMISATION: FIVE STEPS

1. draw it; name the variables
2. the quantity as a function of one variable
3. find the critical points
4. test them, and the ends
5. answer in words, with units